

Human-Computer Interaction — Lecture 02: The Human

المحاضرة 2: الإنسان (The Human)

Multiple Choice (25)

1. Which of the following are listed as the human channels for information input and output?

أي قنوات مذكورة كقنوات إدخال وإخراج المعلومات عند الإنسان؟

- A. Visual, olfactory, gustatory, and thermal
- B. Textual, numerical, graphical, and audio-only
- C. Sensory, short-term, long-term, and working
- D. Visual, auditory, haptic, and movement

✓ الإجابة الصحيحة: D) Visual, auditory, haptic, and movement

ليه صح: المحاضرة بتذكر: visual, auditory, haptic, movement. أنواع الذاكرة (sensory و short-term و long-term) مش قنوات الإدخال/الإخراج.

المرجع: Lecture 02 — Slide 2

2. What is the role of rods and cones in the retina?

ليه دور العصي (rods) والمخاريط (cones) في الشبكية؟

- A. Rods are for colour vision and cones are for low-light vision
- B. Both are only for detecting movement
- C. Rods are for low-light vision and cones are for colour vision
- D. Both focus the image upright

✓ الإجابة الصحيحة: C) Rods are for low-light vision and cones are for colour vision

ليه صح: ال rods للرؤية في الإضاءة الضعيفة وال cones للرؤية الملونة. الصور بتتركز مقلوبة على الشبكية.

المرجع: Lecture 02 — Slide 4

3. Which cells detect pattern and movement according to the lecture?

أي خلايا بتكتشف النمط والحركة؟

- A. Cones
- B. Rods
- C. Ganglion cells
- D. Mechanoreceptors

✓ الإجابة الصحيحة: C) Ganglion cells

ليه صح: ال ganglion cells (مرتبطة بالمخ) هي اللي بتكتشف النمط والحركة. ال mechanoreceptors خاصة باللمس.

المرجع: Lecture 02 — Slide 4

4. What does the visual angle indicate?

إليه اللي بتشير إليه زاوية الرؤية (visual angle)؟

- A. How much of the view an object occupies, relating to its size and distance from the eye
- B. The ability to perceive fine detail
- C. The subjective reaction to light levels
- D. The sensitivity of cones to wavelengths

✓ الإجابة الصحيحة: A) How much of the view an object occupies, relating to its size and distance from the eye

ليه صح: بتشير إلى مقدار ما يشغله الجسم من مجال الرؤية وتتعلق بحجمه وبعده عن العين. القدرة على إدراك التفاصيل هي visual acuity.

المرجع: Lecture 02 — Slide 5

5. Which three attributes make up colour according to the lecture?

إليه الخصائص الثلاثة المكونة للون حسب المحاضرة؟

- A. Hue, intensity, and saturation
- B. Luminance, flicker, and contrast
- C. Pitch, loudness, and timbre
- D. Saccades, fixations, and shape

✓ الإجابة الصحيحة: A) Hue, intensity, and saturation

ليه صح: اللون يتكون من hue و intensity و pitch و saturation و loudness و timbre للصوت.

المرجع: Lecture 02 — Slide 6

6. What design implication follows from the fact that blue acuity is the lowest?

إليه الاستنتاج التصميمي من إن حدة إدراك اللون الأزرق هي الأقل؟

- A. Blue should be used for all important text
- B. Blue is best for low-light displays
- C. Blue should be used to flash warnings
- D. Blue should not be used for important detail

✓ الإجابة الصحيحة: D) Blue should not be used for important detail

ليه صح: لأن blue acuity ضعيف، الأزرق ما ينفعش للتفاصيل المهمة. باقي الاختيارات عكس المنطق.

المرجع: Lecture 02 — Slide 6, 42

7. According to the lecture, what percentage of males and females are colour blind?

حسب المحاضرة، نسبة المصابين بعمى الألوان من الذكور والإناث؟

- A. 1% of males and 8% of females
- B. 50% of males and 25% of females
- C. 0.1% of both
- D. 8% of males and 1% of females

✓ الإجابة الصحيحة: D) 8% of males and 1% of females

ليه صح: 8% من الذكور و1% من الإناث.

المرجع: Lecture 02 — Slide 6

8. Why do optical illusions such as the Ponzo and Muller-Lyer illusions occur?

ليه بتحصل الخدع البصرية زي Ponzo وMuller-Lyer؟

- A. Due to rods failing in low light
- B. Due to damage in the outer ear
- C. Due to overcompensation by the visual system
- D. Due to slow reaction time

✓ الإجابة الصحيحة: C) Due to overcompensation by the visual system

ليه صح: الخدع البصرية بتحصل أحياناً بسبب التعويض الزائد من النظام البصري (للحركة وتغيرات الإضاءة).

المرجع: Lecture 02 — Slide 7, 8

9. In reading, when does perception occur?

في القراءة، إمتى بيحصل الإدراك؟

- A. During saccades
- B. Only at the end of a line
- C. While the eyes are closed
- D. During fixations

✓ الإجابة الصحيحة: D) During fixations

ليه صح: القراءة فيها saccades وfixations، والإدراك بيحصل أثناء الـ fixations.

المرجع: Lecture 02 — Slide 9

10. Which statement about reading from a computer screen is correct?

أي عبارة صحيحة عن القراءة من شاشة الحاسوب؟

- A. Positive contrast always improves reading
- B. Negative contrast improves reading from a computer screen
- C. Word shape is not important to recognition
- D. Colour has no effect on reading

✓ **الإجابة الصحيحة:** B) Negative contrast improves reading from a computer screen

ليه صح: المحاضرة تقول إن negative contrast يحسن القراءة من الشاشة، وإن شكل الكلمة مهم للتعرف عليها.

المرجع: Lecture 02 — Slide 9

11. Which part of the ear transmits sound waves as vibrations to the inner ear?

أي جزء من الأذن ينقل الموجات الصوتية كاهتزازات للأذن الداخلية؟

- A. Middle ear
- B. Outer ear
- C. Inner ear
- D. Auditory nerve

✓ **الإجابة الصحيحة:** A) Middle ear

ليه صح: الأذن الوسطى تنتقل الاهتزازات. الخارجية بتحمي وتضخم الصوت، والداخلية بتطلق مواد كيميائية تسبب نبضات في العصب السمعي.

المرجع: Lecture 02 — Slide 10

12. Which pairing of sound property and physical quantity is correct?

أي ربط صحيح بين خاصية الصوت والكمية الفيزيائية؟

- A. Pitch is amplitude
- B. Pitch is sound frequency
- C. Loudness is frequency
- D. Timbre is amplitude

✓ **الإجابة الصحيحة:** B) Pitch is sound frequency

ليه صح: pitch = التردد، loudness = السعة (amplitude)، timbre = النوع أو الجودة.

المرجع: Lecture 02 — Slide 10

13. What does the cocktail party phenomenon illustrate?

إيه اللي بتوضحه ظاهرة cocktail party؟

- A. Humans hear frequencies above 20 kHz
- B. The auditory system can filter sounds and attend to them over background noise
- C. Hearing is more accurate for high frequencies
- D. Sound is always processed by the middle ear

✓ **الإجابة الصحيحة:** B) The auditory system can filter sounds and attend to them over background noise

ليه صح: النظام السمعي يفلتر الأصوات ويتركز على صوت معين وسط الضوضاء. والبشر يسمعون من 20Hz إلى 15kHz وأقل دقة في الترددات العالية.

المرجع: Lecture 02 — Slide 11

14. Which receptor in the skin detects pain?

أي مستقبل في الجلد بيكتشف الألم؟

- A. Thermoreceptors
- B. Nociceptors
- C. Mechanoreceptors
- D. Photoreceptors

✓ **الإجابة الصحيحة:** B) Nociceptors

ليه صح: nociceptors للألم، thermoreceptors للحرارة والبرودة، mechanoreceptors للضغط.

المرجع: Lecture 02 — Slide 12

15. Which stimulus type has the shortest reaction time according to the lecture?

أي نوع منه له أقصر زمن رد فعل؟

- A. Auditory (about 150 ms)
- B. Visual (about 200 ms)
- C. Pain (about 700 ms)
- D. All are equal

✓ **الإجابة الصحيحة:** A) Auditory (about 150 ms)

ليه صح: السمعي حوالي 150ms مقابل 200ms للبصري و700ms للألم.

المرجع: Lecture 02 — Slide 13

16. What are the three types of memory function described in the lecture?

إيه أنواع وظائف الذاكرة الثلاثة في المحاضرة؟

- A. Episodic, semantic, and procedural memory
- B. Iconic, echoic, and haptic memory
- C. RAM, ROM, and cache
- D. Sensory memories, short-term (working) memory, and long-term memory

✓ الإجابة الصحيحة: D) Sensory memories, short-term (working) memory, and long-term memory

ليه صح: sensory و short-term (working) و long-term. ال episodic و semantic نوعان من LTM، و iconic و echoic و haptic أنواع من sensory memory.

المرجع: Lecture 02 — Slide 15

17. Which sensory memory buffers aural stimuli?

أي ذاكرة حسية بتخزن المنبهات السمعية؟

- A. Iconic memory
- B. Haptic memory
- C. Echoic memory
- D. Semantic memory

✓ الإجابة الصحيحة: C) Echoic memory

ليه صح: echoic للسمعي، iconic للبصري، haptic للمسبي.

المرجع: Lecture 02 — Slide 16

18. What is the limited capacity of short-term memory?

إيه السعة المحدودة للذاكرة قصيرة المدى؟

- A. 70 ± 20 chunks
- B. About 200 chunks
- C. 7 ± 2 chunks
- D. Unlimited

✓ الإجابة الصحيحة: C) 7 ± 2 chunks

ليه صح: 2 ± 7 chunks. زمن الوصول حوالي 70ms والاضمحلال السريع حوالي 200ms.

المرجع: Lecture 02 — Slide 17

19. Which long-term memory type is a serial memory of events?

أي نوع من الذاكرة طويلة المدى هو ذاكرة تسلسلية للأحداث؟

- A. Semantic memory
- B. Episodic memory
- C. Iconic memory
- D. Working memory

✓ الإجابة الصحيحة: B) Episodic memory

ليه صح: episodic للأحداث، و semantic للحقائق والمفاهيم والمهارات، و semantic LTM مشتقة من episodic LTM.

المرجع: Lecture 02 — Slide 19

20. Which model of LTM represents procedural knowledge as condition/action rules?

أي نموذج لل LTM يمثل المعرفة الإجرائية كقواعد شرط/فعل؟

- A. Production rules
- B. Frames
- C. Scripts
- D. Semantic network

✓ الإجابة الصحيحة: A) Production rules

ليه صح: production rules (IF-THEN). ال frames هيكل بيانات بخانات، وال scripts معلومات نمطية لتفسير موقف.

المرجع: Lecture 02 — Slide 24

21. What does the total time hypothesis state?

إيه اللي بتقوله total time hypothesis؟

- A. Learning is best in one single session
- B. Information is lost after exactly 200 ms
- C. Rehearsal is not needed for LTM
- D. The amount retained is proportional to rehearsal time

✓ الإجابة الصحيحة: D) The amount retained is proportional to rehearsal time

ليه صح: كمية ما يُحتفظ به متناسبة مع وقت المراجعة. و distribution of practice effect بتقول التعلم الموزع على الزمن أفضل.

المرجع: Lecture 02 — Slide 25

22. In forgetting, what is retroactive interference?

إيه هو ال retroactive interference في النسيان؟

- A. New information replaces old information
- B. Old information interferes with new information
- C. Information is lost very slowly by decay
- D. Emotion causes subconscious forgetting

✓ الإجابة الصحيحة: A) New information replaces old information

ليه صح: هو أن المعلومة الجديدة بتحل محل القديمة. عكسها proactive inhibition حيث القديمة تتداخل مع الجديدة.

المرجع: Lecture 02 — Slide 26

23. Which kind of reasoning goes from an event to its cause and can lead to false explanations?

أي نوع من الاستدلال يمشي من الحدث للسبب ويمكن يعطي تفسيرات خاطئة؟

- A. Deduction
- B. Induction
- C. Abduction
- D. Analogy

✓ الإجابة الصحيحة: C) Abduction

ليه صح: ال abduction (مثال: لو شفت Sam بيسوق بسرعة افترض إنه سكران). ال deduction بيستنتج منطقياً من المقدمات.

المرجع: Lecture 02 — Slide 33

24. What is the difference between a slip and a mistake?

إيه الفرق بين slip و mistake؟

- A. A slip is a right intention carried out wrongly, while a mistake is a wrong intention caused by incorrect understanding
- B. A slip is a wrong intention and a mistake is a right intention done wrong
- C. Both are caused only by hardware faults
- D. Slips are caused by incorrect mental models

✓ الإجابة الصحيحة: A) A slip is a right intention carried out wrongly, while a mistake is a wrong intention caused by incorrect understanding

ليه صح: slip: نية صح لكن التنفيذ غلط (قلة انتباه أو مهارة). mistake: نية غلط بسبب فهم خاطئ أو mental model.

المرجع: Lecture 02 — Slide 37

25. How does positive affect influence problem solving according to the lecture?

إزاي بيأثر ال positive affect على حل المشاكل؟

- A. It leads to narrow thinking
- B. It leads to creative problem solving
- C. It has no influence
- D. It makes easy tasks harder

✓ الإجابة الصحيحة: B) It leads to creative problem solving

ليه صح: positive affect بيؤدي لحل مشاكل إبداعية، و negative بيؤدي لتفكير ضيق.

المرجع: Lecture 02 — Slide 39

Essay and Calculation Questions (13)

1. Explain the two stages of vision and describe how the eye receives light.

اشرح مرحلتي الرؤية وكيف تستقبل العين الضوء.

Model answer:

Two stages: physical reception of the stimulus, and processing and interpretation of the stimulus.

The eye is a mechanism for receiving light and transforming it into electrical energy; light reflects from objects and images are focused upside-down on the retina.

The retina contains rods for low-light vision and cones for colour vision; ganglion cells (brain) detect pattern and movement.

المرجع: Lecture 02 — Slide 3, 4

2. Explain how the visual signal is interpreted (size and depth, brightness, colour, and compensation).

اشرح كيف يتم تفسير الإشارة البصرية.

Model answer:

Size and depth: visual angle shows how much of the view an object occupies; visual acuity is the (limited) ability to perceive detail; familiar objects are perceived as constant size despite changes in visual angle; cues like overlapping help perception of size and depth.

Brightness: subjective reaction to light levels, affected by luminance and measured by just noticeable difference; acuity and flicker perception increase with luminance.

Colour: made up of hue, intensity, saturation; cones are sensitive to wavelengths; blue acuity is lowest; 8% of males and 1% of females are colour blind.

The visual system compensates for movement and luminance changes; context resolves ambiguity; over-compensation can cause optical illusions such as Ponzo and Muller-Lyer.

المرجع: Lecture 02 — Slide 5-8

3. Describe the stages of reading and the factors that affect it.

اوصف مراحل القراءة والعوامل المؤثرة فيها.

Model answer:

Stages: the visual pattern is perceived; it is decoded using the internal representation of language; it is interpreted using knowledge of syntax, semantics, and pragmatics.

Reading involves saccades and fixations, and perception occurs during fixations.

Word shape is important for recognition, and negative contrast improves reading from a computer screen.

المرجع: Lecture 02 — Slide 9

4. Explain hearing: the ear's physical apparatus, sound properties, and the filtering ability of the auditory system.

اشرح السمع: أجزاء الأذن وخصائص الصوت وقدرة الفلترة.

Model answer:

Hearing gives information about the environment: distances, directions, objects.

Outer ear protects the inner ear and amplifies sound; middle ear transmits sound waves as vibrations to the inner ear; inner ear releases chemical transmitters that cause impulses in the auditory nerve.

Sound: pitch is frequency, loudness is amplitude, timbre is type or quality.

Humans hear 20Hz to 15kHz and are less accurate distinguishing high frequencies than low. The auditory system filters sounds, e.g. the cocktail party phenomenon.

المرجع: Lecture 02 — Slide 10, 11

5. Explain touch and movement, including reaction times and Fitts' Law.

اشرح اللمس والحركة مع أزمنة رد الفعل وقانون Fitts.

Model answer:

Touch provides feedback about the environment and may be the key sense for the visually impaired. Receptors in the skin: thermoreceptors (heat and cold), nociceptors (pain), mechanoreceptors (pressure, some instant, some continuous). Some areas such as fingers are more sensitive. Kinesthesia is awareness of body position and affects comfort and performance.

Movement: response time = reaction time + movement time. Reaction time depends on stimulus: visual ~200 ms, auditory ~150 ms, pain ~700 ms. Increasing reaction time decreases accuracy in the unskilled operator but not in the skilled one.

Fitts' Law: $Mt = a + b \log_2(D/S + 1)$, where a and b are empirical constants, Mt is movement time, D is distance, S is target size. Targets should be as large as possible and distances as small as possible.

المرجع: Lecture 02 — Slide 12-14

6. Describe the three types of memory and the characteristics of each.

اوصف أنواع الذاكرة الثلاثة وخصائص كل منها.

Model answer:

Sensory memory: buffers for stimuli received through the senses (iconic for visual, echoic for aural, haptic for tactile); continuously overwritten.

Short-term memory (working memory): scratch-pad for temporary recall; rapid access ~70 ms, rapid decay ~200 ms, limited capacity of 7 ± 2 chunks.

Long-term memory: repository for all knowledge; slow access ~1/10 second, slow decay if any, huge or unlimited capacity; two types: episodic (serial memory of events) and semantic (structured memory of facts, concepts, skills).

Attention moves information from sensory memory to STM, and rehearsal moves it from STM to LTM.

المرجع: Lecture 02 — Slide 15-19

7. Explain the models of long-term memory: semantic network, frames, scripts, and production rules.

اشرح نماذج الذاكرة طويلة المدى.

Model answer:

Semantic network: represents relationships between bits of information; child nodes inherit properties of parent nodes, which supports inference through inheritance (e.g. Dog, Collie, Beagle, Snoopy).

Frames: information organized in data structures with slots instantiated with values, and type-subtype relationships (fixed, default, and variable slots, e.g. DOG and COLLIE).

Scripts: models of stereotypical information required to interpret a situation, with elements instantiated for context (entry conditions, result, props, roles, scenes, tracks, e.g. a visit to the vet).

Production rules: representation of procedural knowledge as condition/action rules (e.g. IF dog is wagging tail THEN pat dog).

المرجع: Lecture 02 — Slide 20-24

8. Explain how information is stored in, forgotten from, and retrieved from long-term memory.

اشرح تخزين المعلومات في LTM ونسيانها واسترجاعها.

Model answer:

Storage: rehearsal moves information from STM to LTM; total time hypothesis (retention proportional to rehearsal time); distribution of practice effect (optimized by spreading learning over time); structure, meaning and familiarity make information easier to remember.

Forgetting: decay (information lost gradually but very slowly); interference (new replaces old = retroactive interference; old interferes with new = proactive inhibition).

Memory is selective and affected by emotion, so we may subconsciously choose to forget.

Retrieval: recall (reproducing information, assisted by cues such as categories and imagery) and recognition (knowing that information has been seen before; less complex than recall because the information is the cue).

المرجع: Lecture 02 — Slide 25-27

9. Compare deduction, induction, and abduction, and explain the problem solving theories mentioned in the lecture.

قارن بين deduction و induction و abduction و اشرح نظريات حل المشكلات.

Model answer:

Deduction: derive a logically necessary conclusion from given premises (e.g. If it is Friday she goes to work; it is Friday; therefore she goes to work). A logical conclusion is not necessarily true, and people bring world knowledge to bear.

Induction: generalize from cases seen to cases unseen (e.g. all elephants seen have trunks). Unreliable since it can only prove false not true, but useful. Humans are not good at using negative evidence (Wason's cards).

Abduction: reasoning from event to cause (e.g. Sam drives fast when drunk). Unreliable as it can lead to false explanations.

Problem solving: finding a solution to an unfamiliar task using knowledge. Gestalt theory: productive and reproductive problem solving, insight and restructuring. Problem space theory: problem states, legal operators, heuristics such as means-ends analysis, within human information processing limits (e.g. STM). Analogy uses knowledge of a similar problem from a similar domain. Skill acquisition is characterized by chunking and more effective structuring of information.

المرجع: Lecture 02 — Slide 28-36

10. Explain slips and mistakes, the role of emotion in interaction, and why individual differences matter in design.

اشرح الـ slips والـ mistakes ودور العاطفة والفروق الفردية في التصميم.

Model answer:

Slips: right intention but failed to do it right (poor physical skill, inattention; a change to skilled behaviour can cause a slip). Mistakes: wrong intention caused by incorrect understanding; humans create mental models, and if they differ from the actual system, errors occur.

Emotion: theories include James-Lange, Cannon, and Schacter-Singer; emotion involves both cognitive and physical responses. Affect is the biological response to physical stimuli: positive affect leads to creative problem solving, negative affect to narrow thinking. Implications: stress increases problem-solving difficulty, relaxed users are more forgiving, and aesthetically pleasing interfaces increase positive affect.

Individual differences: long term (sex, physical and intellectual abilities), short term (stress, fatigue), changing (age). Designers should ask whether a design decision will exclude a section of the user population.

المرجع: Lecture 02 — Slide 37-41

11. Using Fitts' Law $Mt = a + b \log_2(D/S + 1)$ with $a = 0.1$ s and $b = 0.15$ s/bit, calculate the movement time to hit a target of size $S = 30$ pixels at distance $D = 300$ pixels. Then calculate it again if the target size is doubled to 60 pixels.

[Calculation]

باستخدام قانون Fitts مع $a = 0.1$ ثانية و $b = 0.15$ ، احسب زمن الحركة لهدف حجمه 30 بكسل على بعد 300 بكسل، ثم احسبه لو الحجم اتضاعف لـ 60 بكسل.

Model answer:

Law: $Mt = a + b \log_2(D/S + 1)$.

Case 1: $D/S = 300/30 = 10$; $\log_2(10 + 1) = \log_2(11) = 3.459$; $Mt = 0.1 + 0.15 \times 3.459 = 0.619$ s (about 0.62 s).

Case 2: $D/S = 300/60 = 5$; $\log_2(5 + 1) = \log_2(6) = 2.585$; $Mt = 0.1 + 0.15 \times 2.585 = 0.488$ s (about 0.49 s).

Conclusion: larger targets reduce movement time, consistent with the rule 'targets as large as possible'.

المرجع: Lecture 02 — Slide 14

12. A user responds to a visual stimulus and to an auditory stimulus. Assume the movement time is 300 ms in both cases. Using the typical reaction times from the lecture, calculate the total response time for each. [Calculation]

مستخدم يستجيب لمنبه بصري وآخر سمعي، وزمن الحركة 300ms في الحالتين. احسب زمن الاستجابة الكلي لكل واحد.

Model answer:

Law: response time = reaction time + movement time.

Visual: 200 ms + 300 ms = 500 ms.

Auditory: 150 ms + 300 ms = 450 ms.

The auditory response is 50 ms faster because auditory reaction time (~150 ms) is shorter than visual (~200 ms).

المرجع: Lecture 02 — Slide 13

13. The lecture shows the digit string 212348278493202 and the grouped number 0121 414 2626. Assuming each digit in the first string is one chunk and each group in the second is one chunk, how many chunks is each, and which is within the short-term memory capacity? [Calculation]

المحاضرة بتعرض سلسلة 212348278493202 والرقم المجمع 2626 414 0121. لو كل رقم في الأولى chunk وكل مجموعة في الثانية chunk، كام chunk في كل واحدة وأيهما ضمن سعة STM؟

Model answer:

Capacity of STM = 7 ± 2 chunks (5 to 9 chunks).

212348278493202 has 15 digits, so 15 chunks, which exceeds the maximum of 9, so it is hard to hold in STM.

0121 414 2626 has 3 groups, so 3 chunks, which is within capacity (the lecture's point: chunking helps optimize STM).

المرجع: Lecture 02 — Slide 17, 18